EDUTEX / RC6

Technical Setup and Upgrade Guide

ICT deployment owners

A practical guide for version 1.0.0-rc.6

Prepared from the supplied RC5 project and roadmap. Updated 5 September 2026.

Review build - synthetic data only.
The full roadmap is not complete. All-data client encryption remains unmet; formal ISO assurance and real-environment acceptance are outstanding. Read the scope and release assessment before deployment.

Inside this guide

1. Read this before choosing an environment

This is an RC6 review build based on the supplied RC5 project. It expands the modules and repairs installation, migration, authorisation, accounting and responsive UI defects. It is not a declaration that the entire roadmap is finished or that production acceptance has passed.

Use synthetic records in an isolated evaluation environment. The roadmap's requirement to encrypt all data in the browser before database storage is not met. Selected restricted student fields, clinical notes and child medical records use browser encryption; ordinary operational fields remain readable to authorised database processing. A complete encrypted-record architecture, its migration and its search/reporting design remain release blockers. Disk encryption and HTTPS do not satisfy that requirement.

There is no ISO certification claim. The organisation still needs a defined scope, licensed standards, a control assessment, operating evidence and the applicable independent assessment. Web browsers cannot guarantee that displayed medical information is impossible to photograph or capture. RC6 therefore supplies an online medical view, not the requested guaranteed non-exportable offline app.

Read `docs/RC6_ROADMAP_COVERAGE.md` and `quality/RELEASE_VERIFICATION.md` before planning a live upgrade. Those documents identify implemented behaviour, remaining development and checks that require the real school environment. Preserve the existing RC5 installation until an authorised release decision is recorded.

2. What is easier in RC6

The dependency-free setup entry point gives the operator one consistent command. It checks the computer, installs locked dependencies and starts the existing guarded cloud interview, check, plan and status stages. A separate upgrade interview copies approved school identifiers into a new file and collects a fresh maintenance window. Upgrades skip creation of another tenant or identity pool.

Cloud credentials are not requested by the interview. AWS sign-in uses a named short-lived SSO profile. The Cloudflare provider receives its scoped token only for the commands that need it. Source and plan hashes, private infrastructure, MFA, migration transactions and human plan review remain in place.

Application users need only the school's approved browser address. Node, Docker, Terraform and AWS CLI belong on the ICT operator's machine; parents and teachers do not install them.

3. Architecture and files

Component	Location and responsibility
Public sign-in	`apps/auth-web`; minimal public authentication client
School, parent and student workspaces	`apps/portal-web`; React and Tailwind, account-backed preferences
Technician console	`apps/technician-web`; separately built protected client at `/technician/`
Application API	`apps/api`; Fastify, server sessions, authorisation and workflow transactions
Shared validation	`packages/contracts`; request schemas, record metadata, metrics and alert vocabulary
Database	`packages/database`; migrations 0001-0029, tenant context, PostgreSQL RLS and audit
Infrastructure	`infra/aws` and `infra/cloudflare`; private AWS services and Cloudflare edge
Operator tooling	`scripts`; setup, reviewed deployment, bootstrap and bounded legacy import

The supported deployment remains one school, exact hostname and AWS stack per installation. Cloudflare Tunnel is the public access path. The AWS application and database remain private; do not add a public load balancer, task IP or direct database route to make a failed setup appear healthy.

The ZIP contains maintained source, lockfiles, database migrations, public RDS trust certificates, documentation and verification evidence. Dependencies and build output are regenerated. The UI review fixtures and their simulated accounts are excluded from the deliverable.

4. Prepare the operator computer

Extract the ZIP into a new directory. Open a terminal in `Edutex-production`, the directory containing `package.json`. Keep operator configuration and deployment evidence in a private directory outside that source tree.

Install Node.js 24 or newer, npm 11 or newer, AWS CLI v2, Terraform 1.10 or newer but below 2.0, and Docker Engine 24 or newer with Buildx. Use vendor-supported installers. The Terraform configuration requires Cloudflare provider 5.10 or newer but below 6.0; the exact selected provider lock is bound into the reviewed plan.

```
npm run setup -- check
npm run setup -- install
npm run quality
```

`check` reports missing tools and a stopped Docker engine without changing cloud resources. It is not proof of AWS permissions. `install` runs `npm ci --ignore-scripts`, followed by the specific Sharp rebuild required for photo processing. Do not replace the lockfile with a broad dependency update while following a reviewed deployment.

The strict quality command checks source documentation, types, tests, lint, formatting, all production builds and publication artifacts. Typed linting allows a 6 GiB Node heap; provide sufficient workstation memory. Verify an ARM64 image build on the operator machine because the production task architecture is ARM64. Docker was unavailable in this build environment, so image creation, scanning and signing remain target checks.

The setup launcher avoids shell interpolation and resolves npm on Windows. Windows ACLs, Docker Desktop, installed browsers and cloud deployment still require an actual Windows acceptance test. A successful compilation here does not prove every operating system works.

5. Collect approved cloud values once

Ask the cloud owner to prepare the following worksheet. Record identifiers and ARNs, not secret values, in the setup interview.

Area	Values and prerequisites
AWS	Dedicated account ID, approved region, named SSO profile, required quotas and a ready `CDKToolkit` bootstrap stack
School	Slug, legal name, campus, academic year and dates, IANA timezone, country and initial administrator
Domain	Exact lower-case school hostname, Cloudflare account ID, zone ID and named Tunnel ID
Edge security	Purchased WAF capabilities, TLS policy, approved IP/country policies and administrative networks
Tunnel secret	Same-account, same-region Secrets Manager ARN with an `AWSCURRENT` version containing the actual Tunnel token
Terraform state	Private, versioned S3 bucket in the approved region; customer-managed KMS key ARN; S3 Bucket Key enabled; all public-access blocks enabled
State key	`edutex/SCHOOL_SLUG/cloudflare.tfstate`, using the actual school slug
Email	Verified SES identity ARN, sender address and optional configuration-set name in the same account and region
Alert delivery	Optional verified SES alert sender; SMS disabled unless destination registration, production access and spending controls are approved
Change	Approved change ID, release manager, security approver, start and end timestamps with UTC offsets

Configure AWS SSO using the school's approved permission set, then sign in:

```
aws sso login --profile YOUR_PROFILE
```

The deployment assistant deliberately rejects ambient `AWS_ACCESS_KEY_ID`, `AWS_SECRET_ACCESS_KEY`, `AWS_SESSION_TOKEN`, web identity/role overrides and `TF_VAR_cloudflare_api_token`. Remove conflicting variables using your organisation's terminal procedure and use the named SSO profile.

Create the Cloudflare Tunnel and least-privilege API token through the approved cloud process. Store the Tunnel token directly in Secrets Manager without putting it in source, command history, screenshots or the setup JSON. Supply the provider token as `CLOUDFLARE_API_TOKEN` only in the operator session running the Cloudflare plan/apply. Do not put either token in Terraform input files. Clear the provider token when the work ends.

The current Setup Guide for Everyone retains the detailed numbered cloud and workstation walkthrough. Its Step 17A addresses a missing CDK bootstrap without assuming the absent RC5 companion policy JSON exists. Use approved execution and boundary policies; do not substitute administrator access. The historical manuals remain reference material; use the RC6 assistant for this build. Do not combine an old apply receipt with new source.

6. Create a new-school evaluation configuration

Use a new empty evaluation account/environment and synthetic school details. Obtain valid cloud identifiers first; placeholders do not create usable infrastructure.

```
npm run setup -- configure --output ../edutex-operator/school/rc6.json
npm run setup -- check-cloud --config ../edutex-operator/school/rc6.json
```

The interview validates each answer and the account/region/hostname relationships. Configuration is non-secret but still school-sensitive and is written with owner-only permissions where the operating system supports them. Do not publish it with the source ZIP.

Cloud checks verify local tool versions and Buildx, AWS caller account, CDK bootstrap readiness, Tunnel secret metadata, remote-state controls, KMS and SES prerequisites. They do not retrieve the Tunnel secret value. A passing check is a prerequisite for planning, not a deployment or a production acceptance decision.

7. Upgrade an existing RC5 school safely

Before any live change, close the release blockers, rehearse the upgrade in an isolated restored environment, verify backups and agree a maintenance window. Retain the old ZIP, deployment configuration, receipt, migration checksums, tenant ID, identity pool/client IDs and key/secret ARNs. RC6 migrations are forward-only; an old application ZIP is not a database rollback.

Create the RC6 configuration from the actual RC5 configuration:

```
npm run setup -- upgrade \
  --from ../edutex-operator/school/production.json \
  --output ../edutex-operator/school/rc6.json
npm run setup -- check-cloud --config ../edutex-operator/school/rc6.json
```

The new file has release `1.0.0-rc.6` and purpose `existing-school-upgrade`. It preserves the school's cloud identity and uses fresh change information. It neither overwrites the old file nor reruns new-school bootstrap. Use `configure` only for a genuinely new school.

Perform these data reviews in the restored rehearsal before planning the live change:

1. Resolve duplicate non-null account links in student, staff and guardian records through an approved identity review. Migration 0025 adds uniqueness/category checks and stops if duplicate links remain; it does not delete or merge people automatically.

2. Confirm each guardian account links to the correct guardian record and each student account to the correct student record. Confirm guardians' separate portal, medical, consent and communications rights.
3. Review default parent and student role changes. They receive community portal access only; inherited staff-module grants no longer confer staff access on community accounts. Do not restore broad staff roles as a workaround.
4. Review published forms. Explicit parent and student collection modes separate community forms from internal staff forms.
5. Allocate draft invoice fees explicitly to authorised guardians. Parent fee views show only that guardian's allocation and payments. Existing unallocated invoices do not become visible merely because two people share a family record.
6. Configure school periods, terms, groups and important dates before relying on new calendars or timetable alerts.

The migrator serialises concurrent migration runs and sets audit context for data-changing migrations. Three RC5 extension references required exact source repairs in 0004, 0006 and 0009. `migration-compatibility.ts` accepts only the recorded original/corrected SHA-256 pairs; migration 0012 repairs installed functions. Other changed historical checksums still stop the upgrade. Never edit the migration ledger or invent a compatibility hash to force a run.

The supplied integration suite also upgrades a populated RC5 schema and checks preservation of the existing tenant and administrator. This is useful evidence, but the school's real RDS instance, extension versions and actual data must still be rehearsed.

8. Plan, review, then apply the same bytes

```
npm run setup -- plan --config ../edutex-operator/school/rc6.json
```

Planning runs the quality gate, Terraform validation and saved plan, AWS synthesis and diff. It saves human-readable evidence, the exact Terraform/provider lock and CloudFormation assembly, plus source/configuration hashes. Review the printed evidence paths and `plan-record.json` SHA-256. `PLAN READY` means a plan exists; nothing is approved merely because the command printed it.

Review both cloud diffs with the designated approver. Check school/account/hostname, private networking, KMS and IAM scope, deletion/replacement proposals, desired application count, log retention, SES/SNS access and purchased WAF features. For an upgrade, stop on unexpected replacement of database, identity or cryptographic resources.

Long commands may wrap in the PDF. Use the complete command block from the matching Markdown guide in the ZIP. A final backslash continues that same command onto the next line in Bash or zsh. The following is the explicit apply pattern. Replace every uppercase placeholder with the approved value; do not use a made-up hash.

```
npm run deploy:apply -- \
  --config ../edutex-operator/school/rc6.json \
  --execute \
  --approval-id CHANGE_ID \
  --confirm-hostname SCHOOL_HOSTNAME \
  --confirm-record-sha256 APPROVED_SHA256
```

Use the same `--work-dir` on plan/apply/status if you choose a non-default evidence directory. Keep source, lockfiles, configuration and evidence unchanged between approval and apply. A changed hash requires a new plan and review.

Apply establishes the Cloudflare edge and private AWS foundation with application count zero, runs private migrations, performs school bootstrap only for a new school, then activates the configured service count. It records progress and task ARNs. An existing school experiences a maintenance outage during the zero-task stage; schedule and communicate it.

Check status with:

```
npm run setup -- status --config ../edutex-operator/school/rc6.json
```

Do not create another school when a task fails. Inspect the recorded ECS task ARN and its container exit code/logs. An explicitly failed migration/bootstrap can be retried only through the assistant's exact-task retry controls after correcting its cause. An interrupted operator terminal does not prove that its cloud task failed. Preserve the work directory and use the same reviewed configuration.

9. Authenticate and configure the school

After a successful evaluation deployment, follow the initial administrator invitation through the approved hostname. Complete password/TOTP setup and test sign-out, a new sign-in and session revocation. Keep a tested school recovery procedure and separate authorised recovery administrators.

RC6 creates the Cognito pool with required MFA/password first, configures TOTP and user-verified WebAuthn as a multifactor factor, then enables the passkey first-factor option. This avoids attempting passkey-first MFA before its factor configuration exists. Test passkey registration and sign-in using actual supported authenticators and the managed-login relying-party domain. Current AWS behaviour is described in Cognito MFA documentation.

Microsoft Entra ID, Google Workspace, generic OIDC and SAML configuration remain available. Stage providers disabled; map exact trusted provider claims to narrowly scoped Edutex roles; test allowed, unassigned, disabled and revoked users; then enable the provider. Google identity federation does not import Google Classroom classes, coursework or assignments. Classroom synchronisation remains unimplemented.

Enable modules intentionally and give each role only the work it needs. `manage` covers that module's operations; high-impact operations also require recent MFA and workflow conditions. The technician console requires authorised staff and recent MFA. Parent/student categories cannot acquire staff-module access by receiving a broad staff role.

10. Enter school records in a usable order

Start with campus, academic year, terms, period times, rooms, staff/student/guardian accounts and their verified links. Then create classes and memberships, groups and group managers, and important dates. No-class dates can apply school-wide or to selected year levels. The database prevents conflicting terms and overlapping periods and blocks attendance rolls on applicable no-class dates.

Use the resource search and named reference selectors to link records; staff should not have to type UUIDs. Missing references usually mean the record has not been created, is outside the user's permissions, or is not in the current school. Request the appropriate permission; do not broaden database access.

Configure communications contacts, form collection mode and published learning records before inviting parent and student test accounts. Account-backed dashboard and appearance preferences persist after another device signs in; they are not stored in browser local storage.

For migration from the earlier prototype, the existing `legacy:plan` and `legacy:import` tools remain bounded importers. They are not general accounting or all-module conversion tools. Map and reconcile every source dataset separately, preserve encrypted fields and immutable IDs where required, and review unmapped data. The RC5 handbook describes the existing importer contract.

11. Finance and event acceptance

Create a reviewed chart of accounts (or use the supplied school GL template), budget records and required account references before posting workflows. RC6 supports purchase-order approval, receipt/bill matching, invoice issue, manual payment settlement, balanced journal creation and bank reconciliation. Creator/approver separation, row versions, currency checks and duplicate-source guards apply.

For split fees, create guardian allocations while the invoice is draft. Their sum must equal the invoice total before issue. A settled payment references the appropriate allocation and cannot exceed its remaining share. Test partial payments, repeat attempts and denied access with two guardian accounts from the same family. An unverified browser action does not mean funds were received.

Purchase orders, supplier bills, budgets and payroll use AUD. Invoices, receipts and bank accounts can name a currency, but FX conversion and international payroll are not implemented.

Payroll records support ordinary/overtime minutes, gross/tax review and payment references. Automatic timetable-to-payroll calculations, award interpretation and government lodgement are not supplied. Sibling/scholarship policy records do not automatically recalculate invoices. The 14 fixed reports are review tools; GST output is a working paper, and the balance sheet is explicitly before period closing. Automated card/direct-debit collection, bank transfers, FX, consolidation and a complete statutory accounting engine remain outside this build.

For events, configure venues, risk templates, event templates, approval streams, staff return-to-work considerations, student/staff participation, budget/GL links and emergency arrangements. Template selection copies editable planning defaults. Approvals record decisions and enforce the implemented emergency, supervision and risk conditions; they do not substitute for a staff risk assessment.

Publish eligible event invitations to parents. Staff can generate paper consent forms and record a referenced paper decision in the append-only consent history. Parent portal consent and paper consent remain distinguishable. Use the pending-consent Smart Alert template after previewing its actual time scope.

12. Insights, alerts and delivery

Insights offers KPI, bar and table tiles, colours, selected financial/headcount fields and daily through yearly scopes including terms and a shared dashboard timeline. Choose a currency for monetary tiles. Group managers can share dashboards with groups they manage. The server checks both ownership/sharing and access to the underlying dataset; sharing a dashboard does not grant data access.

Smart Alerts supports ten bounded data sources, all/any conditions, record/distinct-day counts where supported, time windows, dashboard/email/SMS actions and group sharing. Choose recipients explicitly. Save a disabled rule, preview its matches, check the source dates and recipients, then enable it. The engine is not an arbitrary SQL or unlimited automation platform. Its 10,000-row scan guard fails visibly; narrow a large window instead of accepting a partial result.

The worker polls approximately every minute and claims rules on their scheduled interval. Queue deduplication prevents repeated delivery for the same rule/window/matched group/channel/recipient. It rechecks owner permissions. A source privilege revocation stops evaluation or delivery.

Email requires a verified SES sender and production sending permissions. SMS requires SNS configuration, valid E.164 guardian numbers, applicable registrations and approved spend. AWS explains delivery prerequisites in its [SNS SMS overview](). Do not enable live sends in synthetic tests. A provider acceptance receipt is not proof the recipient read the message. Uncertain delivery is retained for operator review instead of automatic repeated sends.

Invoices are evaluated within the chosen source-date window. An older outstanding invoice outside that window will not match; use an appropriate scope and reconcile with the debtor report. Staff SMS numbers and arbitrary external email recipient entry are not supplied by this builder.

13. Medical privacy, retention and access

Restricted student fields and clinical notes use AES-GCM browser envelopes. Parent medical access uses a child-scoped derived key after an explicit relationship check; it does not disclose the school's general field key. Read access is audited. Staff event access requires the configured participant/window conditions; nurse and assigned wellbeing access follow their module/case permissions.

The online medical view clears on hiding, session expiry, record teardown and after its short access deadline (currently 60 seconds). Late key responses cannot restore cleared key material; a new exchange is needed after reopening. Other restricted student/clinical views lock on hiding and after five minutes. Reopen it to reauthorise. Save changes before the deadline. There is no service-worker cache or offline medical download. Print styling suppresses the medical workspace, but a browser cannot stop every screenshot, device capture or camera. Treat the visible endpoint as part of the school's privacy boundary.

Do not put medical narratives into ordinary free-text fields, alerts or form titles. This does not repair the all-data encryption gap: ordinary records, indexes, operational analytics and some audit metadata still depend on server-readable fields. Completing that requirement needs an explicit new cryptographic data model, key lifecycle, existing-data conversion and tests, not a claim that database-at-rest encryption is equivalent.

14. Verification and release decision

Retain the RC6 quality log and dependency audit with the build. The local integration suite uses real PostgreSQL through PGlite, applies all 29 migrations, switches to the restricted application role and tests workflow transactions and negative authorisation. Its HTTP identity is a test adapter; it does not prove Cognito/RDS/IAM behaviour in AWS.

The browser review used synthetic records in Chrome at 320, 390 and 768 pixel widths and desktop width. It inspected the dashboard, appearance, navigation, tables, insights, alerts, family/student dialogs and technician client. Physical iPhone/Android, Safari, Firefox, Windows/macOS/Linux combinations, keyboard/screen-reader audits and installed home-screen behaviour remain acceptance checks. The final repeat browser pass was blocked by the review browser URL policy; the final lifecycle fixes were checked with the compiler and cryptographic regression tests.

Before admitting client data, close the documented development blockers and run the actual cloud plan, Docker build/scan/signing, RDS migration rehearsal, two-tenant and guardian isolation tests, identity/factor recovery tests, WAF/direct-origin tests, SES/SNS tests, load tests, isolated backup/restore and independent security assessment. Keep business, privacy and clinical owners involved in their own acceptance scenarios.

If a migration or acceptance check fails, keep the application unavailable to ordinary users, preserve the logs and investigate. Restoring a database is a controlled recovery into an isolated verified environment with its matching application and key material. Do not drop new tables, edit migration checksums or delete audit history as an improvised rollback.

15. Troubleshooting and handover

Symptom	Next step
`node`/`npm` not found	Install the supported runtime, reopen the terminal and rerun the computer check.
Docker needs attention	Start the engine, confirm Buildx and ARM64 capability; rerun the check.
Wrong AWS account or credentials rejected	Sign in with the approved SSO profile; remove conflicting static credential variables.
State/KMS/Tunnel preflight failure	Correct the named prerequisite resource or policy; never disable the check.
Plan hash mismatch	Recreate and review a plan from the final source/configuration.
Migration checksum mismatch	Compare the exact release file and recorded ledger; only the three documented RC5 repairs are accepted.
Duplicate person-account link	Correct identity links in the restored rehearsal under approved data stewardship; no automatic merge.
403 or recent-MFA prompt	Verify account category, module permission and fresh sign-in.
Parent has no children or fees	Verify the person-account link, guardian rights and explicit invoice allocations.
Medical view closes	Reopen online and reauthorise; it intentionally does not retain offline content.
Alert produces no match	Check source permission, selected time window, condition types and preview output.
Alert is blocked/uncertain	Inspect the delivery record and provider evidence before considering a resend.
Record changed conflict	Reload, compare the latest row and reapply only the intended edit.

Hand over the exact ZIP checksum, configuration and plan/receipt paths, approved cloud inventory, recovery contacts, key/secret inventory, backup evidence, role matrix, open roadmap items and support ownership. Keep credentials outside this handover document.